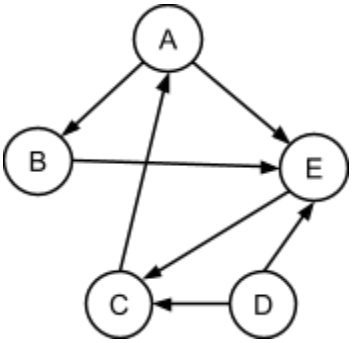
# American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Intermediate Division

**1. Graph Theory**

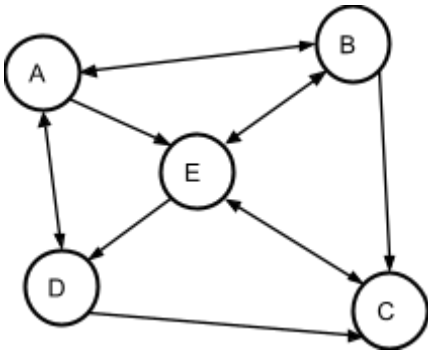
How many paths of length 2 are there in the following directed graph?




A. 6
B. 7
C. 8
D. 9
E. 10

**2. Graph Theory**

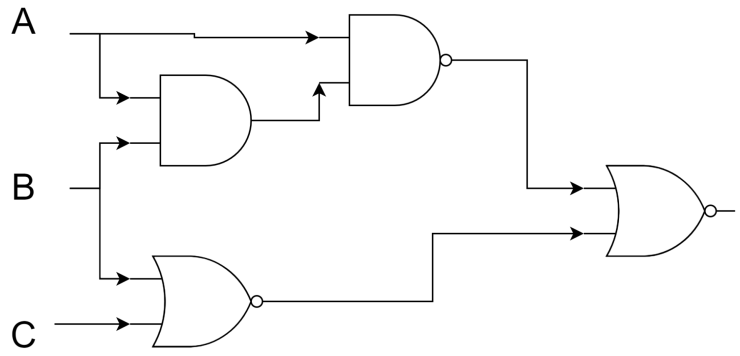
How many cycles are there in this directed graph?




A. 8
B. 9
C. 10
D. 11
E. 12

**3. Digital Electronics**

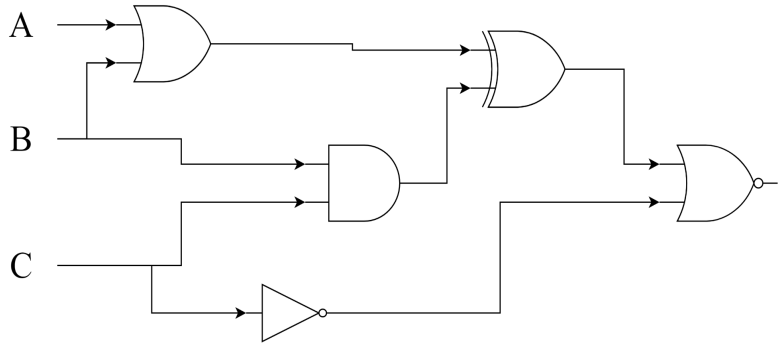
How many ordered triples make the following digital circuit TRUE?




A. 0
B. 2
C. 4
D. 6
E. 8

**4. Digital Electronics**

Translate the following digital circuit to a Boolean algebra expression.




A. $\overline{AB \oplus (B + C) + \overline{C}}$
B. $\overline{(A + B) \oplus BC} + \overline{C}$
C. $\overline{A + B \oplus BC + \overline{C}}$
D. $\overline{(A + B) \oplus BC + C}$
E. $(A + B) \oplus BC + \overline{\overline{C}}$

**5. Assembly Language**

What is the output when the following program is executed?

```
NUM     DC      2025
        LOAD    NUM
        DIV     =100
        STORE   A
        MULT    =100
        SUB     NUM
        MULT    =-1
        STORE   B
        DIV     =5
        MULT    =10
        STORE   ANS
        LOAD    A
        DIV     =5
        ADD     ANS
        STORE   ANS
        PRINT   ANS
        END
```

A. 45
B. 54
C. 405
D. 504
E. 99

**6. Assembly Language**

What is the output when the following program is executed?

```
T       DC      0
N       DC      19
NUM     DC      135
TOP     LOAD    NUM
        SUB     T
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

A. 21
B. 22
C. 23
D. 24
E. 25